VedaVenture

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Abstract

The traditional methods of learning arithmetic operations emphasize procedural sophistication and drilling. This leads to a lack of motivation and retention of students. However, with mobile learning and gamification becoming very popular for improving learning motivation by making learning accessible anytime and anywhere, there is still a lack of association with highly efficient computing frameworks. To address this issue, a complete mobile learning app was developed using Android Studio. This app has the potential to transform the mental training of students in math by educating and training them on Vedic Mathematics. Vedic Mathematics uses simple rules and patterns to simplify the process of computing. This makes it highly efficient for fast mental computation. This app integrates these highly efficient methods of mathematics with gamification and mobile learning space. A secure login system for protecting user data and elements of gamification and mlearning space are incorporated. These elements include points, learning levels, rewards, and learning streaks. Finally, with gamification and mLearning space incorporated with highly efficient computing methods, this app offers an incredibly interactive platform for improving learning and computation of students.

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Keywords: mLearning, Vedic Math, Gamification, Adaptive Learning, Personalization

1 Introduction

Mathematics plays a fundamental role in education, forming the basis for problemsolving, logical reasoning, and analytical thinking. However, many learners struggle with traditional arithmetic methods due to their procedural complexity and lack of engagement. Conventional teaching approaches often rely on repetitive practice and memorization, which can lead to reduced motivation and poor retention among students.

In recent years, digital learning technologies have emerged as effective tools to enhance educational experiences. Among these, gamification has gained significant attention for its ability to improve user engagement and motivation by incorporating game design elements such as points, levels, and rewards into nongame contexts [**?**]. Studies have shown that gamified learning environments can increase participation and enhance learning outcomes, although their effectiveness depends on proper design and user context [**?**].

Simultaneously, mobile learning (Mlearning) has transformed the accessibility of education by enabling learners to access content anytime and anywhere. The widespread adoption of smartphones has made mobile platforms a powerful medium for delivering interactive and personalized learning experiences [**?**]. Mobilebased systems are particularly suitable for short, engaging, and practiceoriented activities such as arithmetic training.

Another important advancement in educational technology is the development of adaptive learning systems. These systems personalize the learning experience by adjusting content difficulty and feedback based on individual user performance. Personalized gamification has been shown to significantly improve motivation, engagement, and cognitive outcomes compared to generic approaches [**?**].

In parallel, Vedic Mathematics offers an alternative approach to conventional arithmetic by providing efficient mental calculation techniques based on simple rules and patterns. These methods reduce computational complexity and improve calculation speed and accuracy, making them highly suitable for rapid problemsolving [**?**]. Despite its advantages, Vedic Mathematics has not been widely integrated into modern digital learning platforms.

Although existing research has explored gamification, mobile learning, and adaptive systems independently, there is limited work that combines all these approaches specifically for mental arithmetic training. Furthermore, most current applications lack integration with efficient computational techniques such as Vedic Mathematics, resulting in missed opportunities to enhance both engagement and performance.

To address these limitations, this project proposes a gamified, adaptive mobile application for learning Vedic Mathematics. The system incorporates interactive challenges, dynamic difficulty adjustment, and rewardbased progression to create an engaging and effective learning experience. By combining gamification, personalization, and efficient mathematical techniques, the proposed solution aims to improve both user motivation and computational proficiency.

The main contributions of this work are as follows:

- Development of a gamified mobile application for arithmetic learning.

- Integration of Vedic Mathematics techniques for efficient computation.
- Implementation of adaptive difficulty based on user performance.
- Enhancement of user engagement through streaks, rewards, and interactive challenges.

2 Literature Review

2.1 Gamification in Education

Gamification is increasingly being successfully used as a strategy for improving user interactions and motivating motivation from outside of gaming contexts. According to Deterding et al., gamification refers to the incorporation of gamelike elements (points, badges, and leaderboards) into game designs for nongaming purposes. Thus the modifications made to create gameful experiences and the concepts developed within this initial body of research laid the foundation for development of theoretical constructs for investigating the implementation of gamified systems. Yet there have been few empirical investigations into gamified systems to support the validity and reliability of such research. Hamari et al. provided a review of many of the empirical studies on gamification and concluded that gamification generally creates increased user participation, engagement, and enthusiasm. The acknowledgement of the situational specificity of the effectiveness of gamification and user characteristics in determining the effectiveness of gamification was a significant finding of this research. Furthermore, Ivankovi c et al. created an adaptive, gamified learning system called Codey to teach coding. Codey included features such as experience points (XP), streaks, leaderboards, and varying levels of complexity. Evidence of the increased interaction between students and improved learning outcomes among students was demonstrated. However, there is still a lack of large scale validation for Codey's findings.

Table 1: Comparison of Existing Literature

Author (Year)	**Focus Area**	**Methodology**	**Key Findings**	**Limitations**
Deterding et al. (2011)	Gamification Theory	Conceptual study	Defined gamification and game elements	No empirical validation
Hamari et al. (2014)	Gamification Review	Literature review	Gamification improves engagement	Contextdependent results
Ivanković et al. (2025)	Gamified Learning System	System development + testing	Improved engagement using XP, streaks	Limited sample size
Xiao & Hew (2024)	Personalized Gamification	Experimental study	Personalization improves motivation	Requires complex user modeling
Crompton (2013)	Mobile Learning	Historical analysis	Enables flexible learning	Lacks implementation focus
Patel & Parekh (2021)	MLearning Adoption	Surveybased study	High mobile usage in education	Limited scalability discussion
AlFahad (2009)	User Perception	Survey analysis	Positive student perception	Device limitations
Träff et al. (2025)	Arithmetic Learning	Longitudinal study	Cognitive + number skills important	Focused on children only
Dowker (2019)	Cognitive Differences	Theoretical + empirical insights	Individual differences affect learning	Not appbased
Nawale et al. (2026)	Vedic Mathematics	Experimental analysis	Faster and accurate calculations	Limited digital integration

2.2 Personalized and Adaptive Learning Systems

The recent studies have emphasized the role of personalization in gamified learning. Xiao and Hew explored the concept of the personalized gamification and contrasted it with the onesizefitsall strategies of the past era of research

Adaptive learning systems also extend the concept of personalization and dynamically vary content and difficulty in response to user performance. Some other adaptive mechanisms, like the feedback loops and difficulty scaling, were also implemented in the Codey system, which allows the learners to study at their pace

2.3 Mobile Learning (MLearning)

By allowing for learning at any time and in numerous locations, Mlearning has altered the face of education (Crompton). Crompton has demonstrated the evolution of mlearning from past forms of education into learnercentered education. In this study, Crompton also discusses the aspect of accessibility and flexibility through the use of mobile platforms

Patel and Parekh evaluated mobile learning in colleges and universities. They report that because of the widespread use of smartphones by the majority of university students and faculty, access to educational materials has been increased tremendously. Based on these results, the researchers concluded that all mlearning systems will need to be optimized for ease of use and student engagement in order to achieve maximum effectiveness.

Subsequent to the Patel and Parekh study AlFahad investigated the attitudes of students toward mobile learning (Mlearning) as a tool for enhancing their education AlFahad found that most learners viewed mobile learning positively for both its convenience and communication capabilities, but there were some challenges associated with the use of mobile devices for learning, particularly related to the limitations of their devices and the potential for distraction.

2.4 Mathematics Learning and Cognitive Factors

The cognitive aspects of learning mathematics need to be understood to develop effective learning tools Träff et al. studied mental arithmetic skill development and concluded that both domainspecific number skills and domaingeneral cognitive abilities contribute to arithmetic skills Their study also revealed that learning and performing require different cognitive processes.

Dowker studied individual differences in arithmetic abilities and emphasized how arithmetic performance varies significantly from person to person due to cognitive, emotional, and experiential factors

2.5 Vedic Mathematics and Computational Efficiency

Vedic Mathematics offers an alternative approach to traditional arithmetic methods by using simplified techniques for faster computation. Nawale et al. demonstrated that Vedic Mathematics significantly improves computational speed and accuracy by

reducing cognitive load and simplifying calculations [**?**]. The study also highlighted its applicability in modern domains such as engineering and computer science.

The integration of Vedic Mathematics into modern educational systems has the potential to enhance problemsolving skills and promote efficient mental calculation techniques. However, its adoption in digital learning platforms remains limited, indicating a gap in current research.

Table 2: Feature Comparison of Existing Approaches

System / Study	Gamification	Personalization	Mobile Support	Adaptive Difficulty	Math Focus
Deterding et al.	Yes	No	No	No	No
Hamari et al.	Yes	No	No	No	No
Codey System	Yes	Yes	No	Yes	No
Xiao & Hew	Yes	Yes	Yes	Yes	No
MLearning Studies	No	No	Yes	No	No
Arithmetic Studies	No	Yes	No	No	Yes
Vedic Math Study	No	No	No	No	Yes
Proposed System	Yes	Yes	Yes	Yes	Yes

2.6 Summary and Research Gap

According to the literature, gamification, personalization, and mobile learning are highly effective in improving the outcome of education. Although gamification can be used for better engagement and motivation, personalization can be used for better results by considering the differences between individuals. Mobile learning can also be used for better accessibility, making it a suitable option for modern education.

However, existing systems are mostly based on general gamification techniques, and efficient mathematical techniques such as Vedic Mathematics are not considered in these systems. Moreover, research on adaptive gamification with mobile devices for mental arithmetic training is still in its infancy. This project proposes a gamified adaptive mobile application that uses Vedic Mathematics for better computational efficiency and engagement.

Table 3: Identified Research Gaps

Existing Limitation	Proposed Solution
Onesizefitsall gamification	Personalized gamification using user preferences
Lack of mobilefirst systems	Develop mobilebased learning application
No integration of Vedic Mathematics	Incorporate Vedic techniques for fast calculation
Limited adaptive difficulty systems	Dynamic difficulty adjustment based on performance
Lack of engagement tracking	Use streaks, XP, and rewards system

3 Objectives

The primary objectives of VedaVenture are as follows:

1. Create an educational mobile application that will teach students about Vedic Maths in a structured manner through gamification and step-by-step learning on an Android device.
2. Develop a Career Mode (helps the student understand the concepts of Vedic Mathematics) and Challenge Mode (the student will race against time and their friends to solve as many problems as possible).
3. Use an Optical Character Recognition (OCR) engine to scan real-world arithmetic problems, identify the associated Vedic Sutra, and provide step-by-step guidance to the user to solve the problem.
4. Create an adaptive difficulty mechanism that will adjust the difficulty of questions based on the user's performance metrics such as accuracy and speed of response.
5. Implement a complete gamification and engagement framework incorporating XP (Experience Points), daily streaks, achievement badges and level progression to keep users engaged over the long term.
6. Use a method for unveiling new Sutras that will allow the user to master the concepts behind the Sutra before they can move on to the next Sutra. This will prevent users from skipping foundational techniques.
7. Include hints to questions in Career Mode to assist the user with their learning, and to reduce cognitive overload when the user has difficulty completing the question.
8. Use Firebase Authentication and Cloud Firestore to create user accounts that are secured and available on multiple devices.
9. Use the principles of Material Design to create an accessible and inclusive mobile interface for users of all ages and abilities, and to optimize the user experience across all screen sizes.
10. Show the educational technology industry that it is possible to combine ancient methodologies of mathematics with modern mobile learning platforms that use gamification.

4 Methodology

4.1 System Overview

The proposed system is a mobilebased Vedic Mathematics learning system, which can improve users' arithmetic skills. The system is based on a modular and serviceoriented architecture, using an Androidbased frontend and Firebase Authentication and Cloud Firestore for the backend.

The system is composed of a user interaction component, learning and challenge parts, common computational parts, and a cloudbased database for storage.

4.2 System Architecture

The system architecture is composed of three main layers. The frontend client, which is based on Android, is responsible for user interaction, providing realtime feedback, and using a joystick for input. The second layer is composed of the functional layer, which contains the Practice Module and Challenge Module. The backend layer utilizes Firebase Authentication for user management and Firestore for storage.

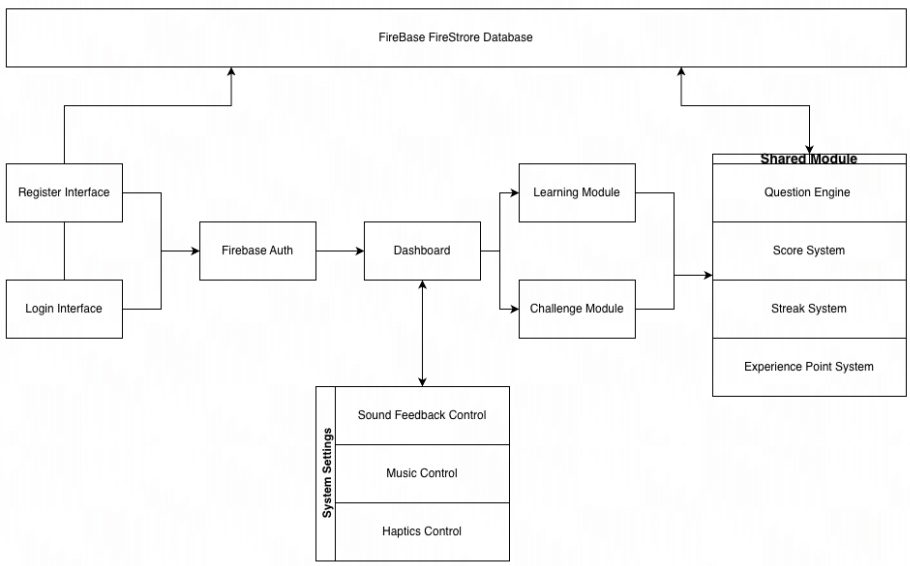


Fig. 1: System Architecture of the Proposed Application

4.3 Overall System Workflow

As depicted in Figure 2, the system starts from user authentication through the login or registration process. Once the user successfully authenticates the account, the system will redirect the user to the dashboard where the user can select between the Practice Module and the Challenge Module. Other features are also incorporated into the system.

4.4 Practice Module Workflow

The Practice Module is for conceptual understanding of Vedic Sutras. The workflow of this system starts from choosing a Sutra based on user progress. The system will demonstrate, and then users can solve stepbystep problems. The response of users will be checked, and feedback will be generated. If it is wrong, users can try again. After correct completion, it will be updated in the database.

4.5 Challenge Module Workflow

The Challenge Module uses gamification through timebound problem solving. The questions are generated dynamically depending on the difficulty level. The user input is taken through a joystick. The accuracy of the input and the updating of the score, experience points, and streak are checked. The process continues until the time expires. Then, the results are computed and saved.

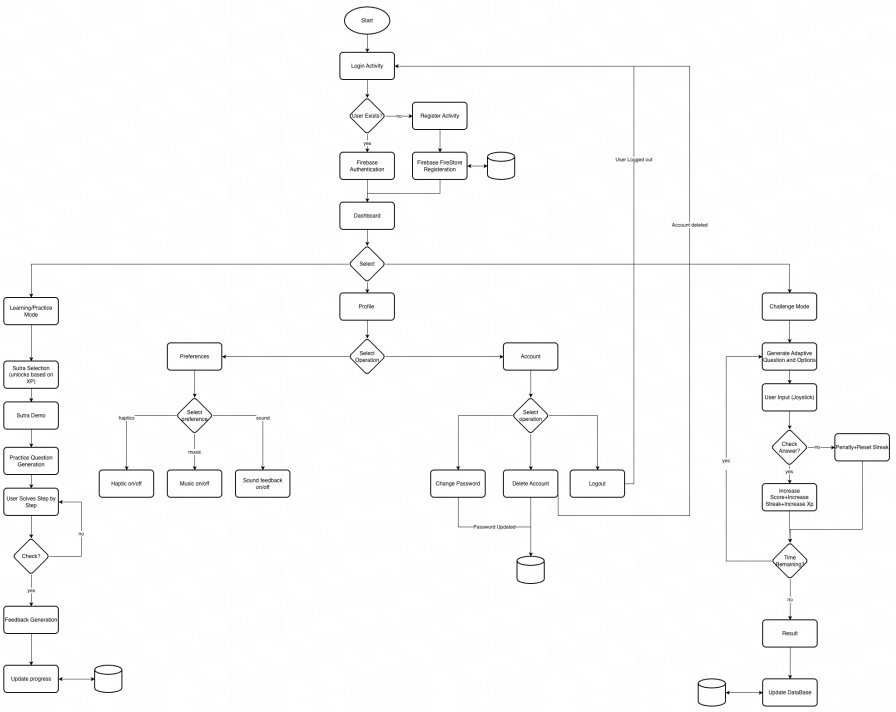


Fig. 2: Overall System Workflow of the Proposed Vedic Mathematics Application

4.6 Adaptive Difficulty Mechanism

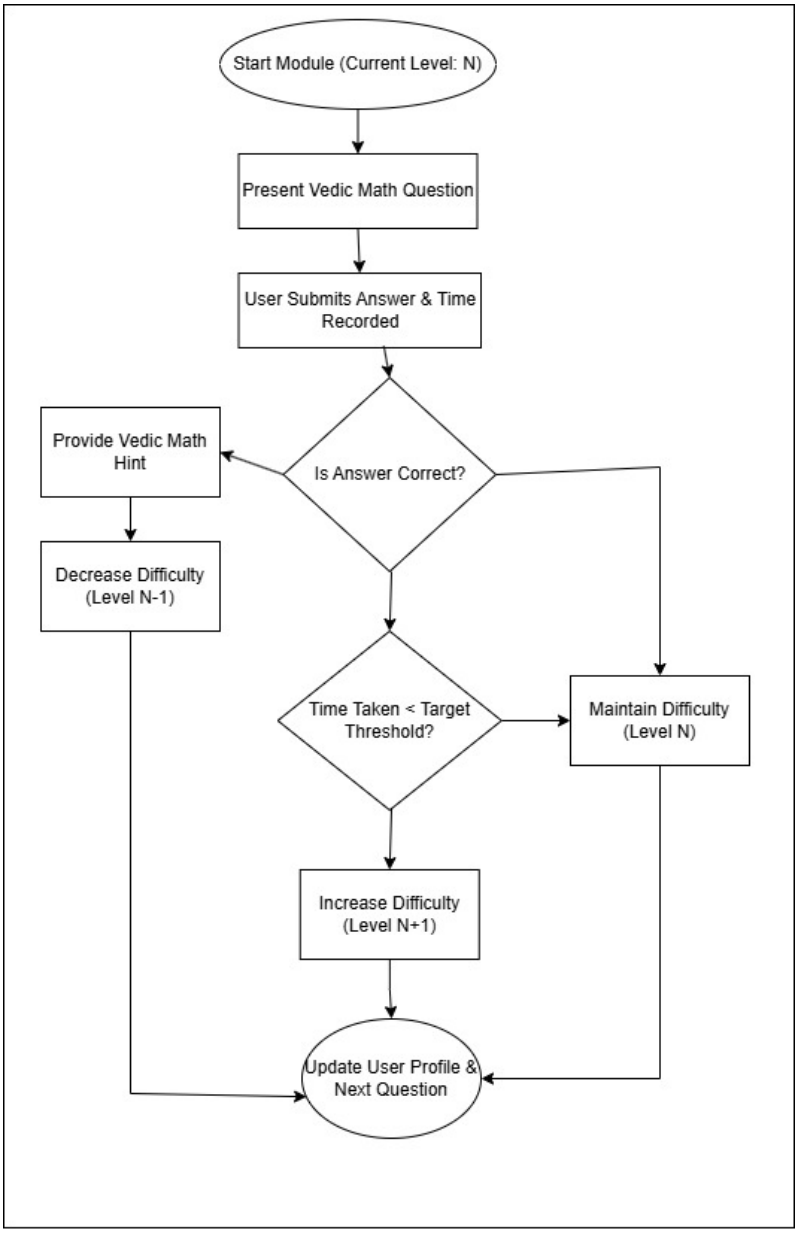


Fig. 3: Adaptive Difficulty Adjustment Mechanism

It also has an adaptive difficulty level that depends on the user's performance, as illustrated in Figure 3. The difficulty level increases if the user correctly answers the question within the desired time, remains the same if the answer is correct but given at a slower rate, and decreases if the answer is incorrect.

4.7 Question Generation Strategy

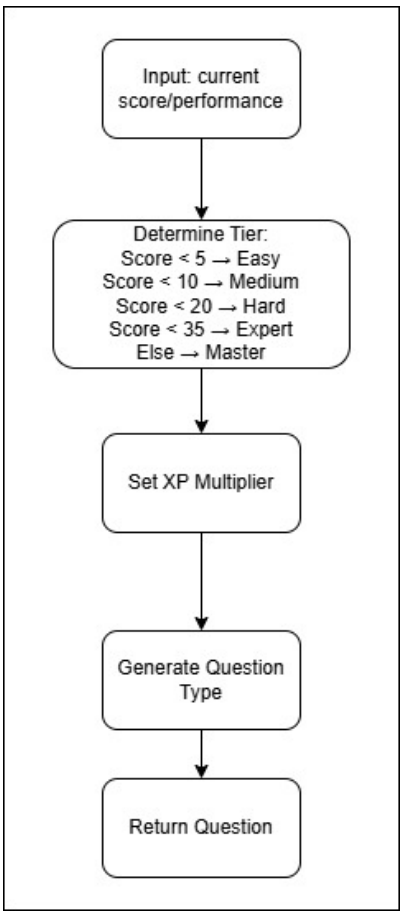


Fig. 4: Question Generation Process Based on Difficulty Levels

The question generation module identifies the difficulty tier depending on the user's score, as shown in Figure 4. Each difficulty tier has a corresponding XP multiplier and question type. Vedic arithmetic questions are generated according to this.

4.8 JoystickBased Input Handling

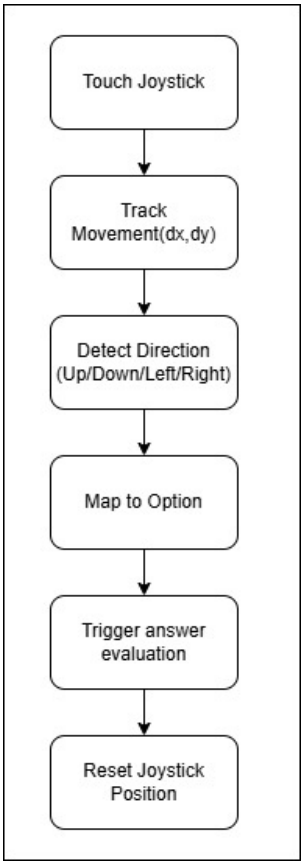


Fig. 5: Joystick Input Processing Mechanism

It (5) uses a joystick input mechanism to improve interactivity. The input mechanism works as follows. The user moves their touch input, and this is translated into a direction. The directions are used as answer choices. The answer choice is then evaluated, and the joystick returns to its original position.

4.9 XP and Streak Management

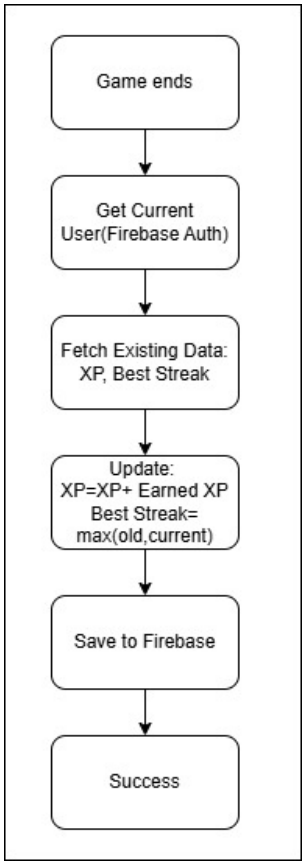


Fig. 6: XP and Best Streak Update Process

The gamification system keeps track of experience points (XP) and the best streak for the user. At the end of each session, it retrieves all the data from Firestore for the user, as shown in Figure 6. The experience points are incremented by the earned value and the streak is updated as the maximum of the previous best streak and the new streak. The new values are stored .

4.10 Data Persistence and Cloud Integration

Firebase Firestore is used for realtime storage and synchronization of data. The database stores user credentials, XP, streaks, and performance information. Firebase Authentication is used for secure management of user accounts. This approach offers scalability, reliability, and simplicity in backend management.

4.11 System Integration

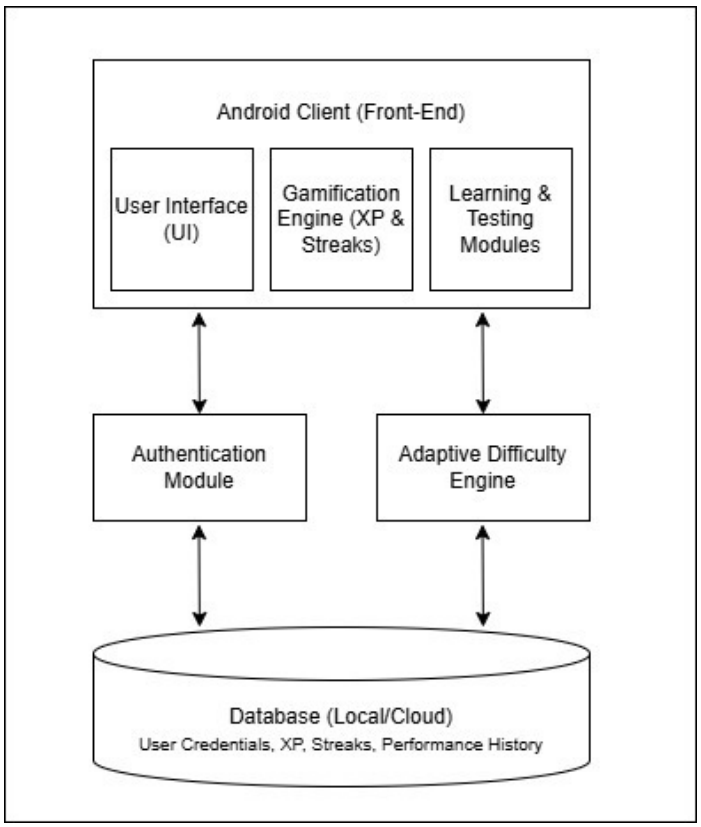


Fig. 7: Abstracted System Integration Model

As a whole, the system comprises different components, which are the user interface, adaptive difficulty, gamification, and database (firebase firestore), as shown in Figure 7.

Presentation Layer (FrontEnd): In this layer, the user interface was designed using Extensible Markup Language (XML) for faster response and better accessibility for different screen sizes of the mobile devices used by the users. The activities of the android are responsible for handling the main states of the application, which are login and dashboard. In addition, the application uses a fragment for content switching while a user is either learning or testing, as opposed to reloading the screen.

Application Logic Layer (Middleware): In this layer, the logic of the application was designed using Java, based on ViewModel and LiveData of the android operating system, for observing the change of data in realtime, ensuring that as soon as the user's scores or the number of wins in a row are updated in the database, they are reflected on the user interface immediately.

Data Management and Security: To manage user credentials and personalized data in a secure manner, the system utilizes Firebase Authentication for user verification. The passwords and credentials are managed securely by the Firebase Authentication mechanisms. To manage the data, the system utilizes the Cloud Firestore database for storing userrelated data, including experience points (XP), streaks, and performance metrics. The data is stored in the cloud, allowing realtime synchronization of the data, thereby eliminating the need to store the data locally using storage mechanisms.

4.12 OCR Scanner Module

VedaVenture has one of its most unique characteristics in that it includes an Optical Character Recognition (OCR) scanner as part of its module set so that when any user points their device's camera at any printed or handwritten arithmetic problem, the application will automatically scan and identify both the question, and the Vedic Sutra that is best suited to solve that question, with a full breakdown of how to achieve the final solution being provided by VedaVenture.

The OCR pipeline operates in the following manner: from either the main Dashboard or another module that allows access, an OCR scanner will be activated by the user. At that point, the real time camera feed from the device will be displayed through the application. The user will take a picture of the arithmetic question. The OCR engine will process the scanned image to read and extract the textual content of the arithmetic expression contained within the image. The pattern matching and classification module will process the extracted arithmetic expression to determine what type of arithmetic expression it is (e.g. multiplication squaring, addition close to a base, etc.) and determine the Vedic Sutra that is best suited to solve that type of arithmetic expression; The results will be displayed in a similar format to that of the Career Mode

The advantage to this feature is that it enables students to see a connection between their classroom learning and real life, and students can therefore use VedaVenture as a resource for completing homework assignments, preparing for exams, and/or solving arithmetic problems in general. As a result, the application will have increased functionality outside of just practicing inside the application, by extending the utility and scope of VedaVenture to solve real world math problems.

4.13 Integration of Vedic Mathematics Modules

The educational portion at the app involves converting the traditional Vedic math formulas and rules – like Urdhva Tiryakbhyam (Vertically and Crosswise) and Nikhilam Navatashcaramam Dashatah (All From 9's and Last From 10's) – into digital form to help people easily multiply, and square, large numbers. Each educational module consists of 2 major components:

Instructional Component Users will be introduced to a new Vedic math rule through the methodical breakdown of the computation with visual animation in the instructional component of the module. by using the methodical visual breakdown of the computation process, users will have a better chance of reducing the cognitive overload that often accompanies learning a new technique.

Application Component Users will interactively engage in time-limited activities to apply what they've learned from the instructional component of the module in order to bridge the gap between the learning and the speed in which the user executes the computation in their mind.

Each Vedic Sutra will be transformed into a structured learning module within each users' Career Mode. The process of converting each Vedic Sutra into a digital learning module includes converting each of the Sutras' rule into a defined algorithm with clearly defined steps. Each step will have a visual demonstration that will use watching someone work through the numeric example as a means of demonstrating to the user how to utilize the sutra in their computations. The instructional design flow for the instructional portion of the module will include first introducing the user abstractly to the Vedic Rule, then demonstrating a concrete and visual demonstration of the Vedic Rule, and finally challenging the user to a practical application of the Vedic Rule on their own.

4.14 Adaptive Difficulty Algorithm

In order to cater to the various learning needs and maintain the optimal state of engagement, a dynamic adaptive difficulty engine has been engineered. The user performance is assessed on two key factors accuracy and response time.

The performance score for the user is computed by the engine after the user completes the question. In the case where the user is able to solve the problem correctly within a certain threshold (for example, 80% accuracy and an average response time of ≤ 5 seconds), the difficulty is increased by providing the user with more digits and/or applying advanced Vedic math techniques.

On the contrary, if the user's accuracy is less than a certain threshold, the difficulty is decreased by providing the user with certain hints and/or reviewing the relevant Vedic Sutra.

4.15 Gamification and Retention Engine

In an effort to compensate for the lack of motivation that is normally present in the traditional arithmetic drilling approach, the entire gamification system was incorporated.

Experience Points (XP) and Level Progression: The user is rewarded with various amounts of XP depending on the pace and accuracy of their response. By accumulating XP points, the user can move through various levels of the entire learning curriculum.

Daily Streak: A system was implemented to encourage the user to develop the habit of learning every day. By completing at least one mathematical problem, the user can increase the daily streak counter. A lack of activity for 24 hours resets the counter to zero.

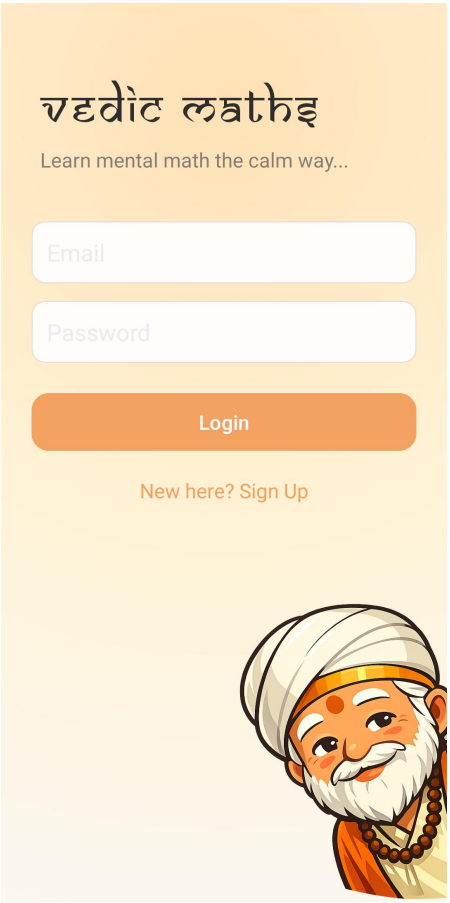
Reward System: The user is rewarded by receiving badges conditionally upon the completion of certain milestones, such as "Master of Multiplication" and "7Day Streak".

5 Results

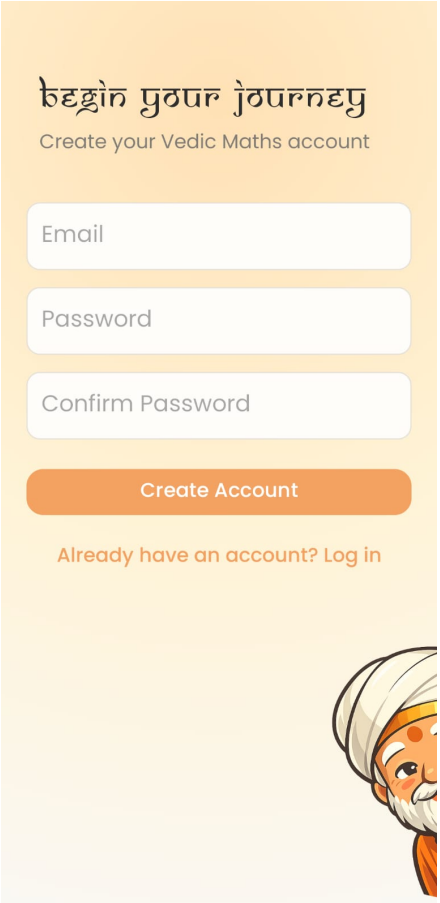
The application that we developed presents a clean, Material Designcompliant interface that guides wonderfully users through their learning journey. The following sections describe the key screens and their roles in the user experience.

5.1 Login and Registration

The Login screen allows existing users to enter their registered email and password to access the application. In case of invalid username or password, appropriate error messages are displayed to the user. Firebase Authentication handles credential verification, and a successful registration redirects the user directly to the Dashboard. Upon successful authentication, users are redirected to the main Dashboard, where they can continue their learning journey from their saved progress.



(a) Login Screen



(b) Registration Screen

Fig. 8: Authentication Screens

5.2 Dashboard

The primary navigation page of the this app is called the Dashboard, which links users to quickly check their learning accomplishments, core modules and profile functionalities. The dashboard is visually designed to have an engaging yet minimal look to illustrate the educational/gamified nature of the application.

On the upper part of the dashboard screen, there are key performance indicators that display the user's total experience points (XP) and the user's best streak. The purpose of these indicators is to encourage practice consistency and allow the user to track their long term level of accomplishments. There are also quick access icons that provide immediate access to the assorted functions including the OCR scanner and personal profile/setting

The center area of the dashboard will feature the theme branding of the application, along with a character depiction, which will provide a bright and compelling identity to the user interacting with the application. This design will enhance user input and allow for more recurrent use of the application, particularly for younger learners.

Two primary action buttons are presented prominently:

- **Continue Journey:** Redirects the learner to Career Mode, where they can continue structured lessons, unlock new Sutras, and progress through guided learning modules.
- **Challenge Mode:** Opens the competitive practice environment where users solve timed arithmetic problems based on their current skill level to earn XP and improve streaks.

The dashboard is intentionally optimized for fast navigation, ensuring that users can begin learning or practicing within a single tap. It also serves as a personalized home screen by continuously updating user progress data through Cloud Firebase Firestore sync.

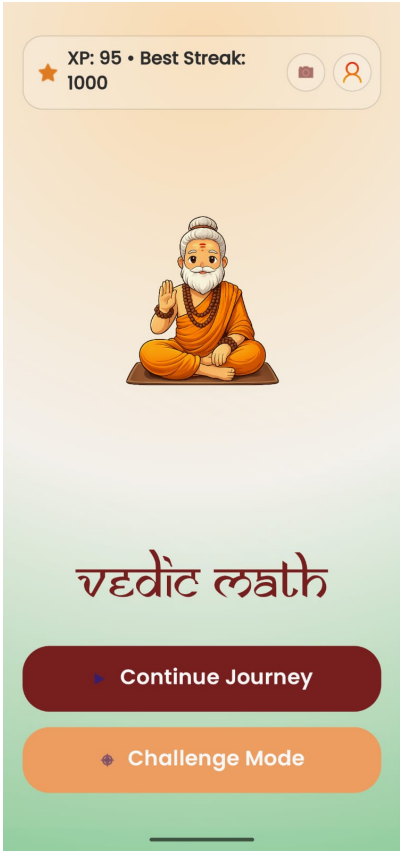


Fig. 9: Dashboard Screen

5.3 Settings and Profile Page

Users can manage their account information, track their accomplishments, and customize their in-app experience from one central location with the Settings and Profile Page. The Settings ad Profile Page combines profile analytics through the Analytics page in the app and preference settings into one streamlined view. This is an important part of both the overall usability of the application and the ability to maintain long-term engagement with the application

The top of this page is the user's profile section. This is where the user sees their registered email address, avatar, and their cumulative Total XP. By presenting this information so clearly, it helps create a sense of ownership for the user and provide a visual cue for how far they've progressed in the application; thus, encouraging them to stay active on the platform.

The **Preferences** section allows users to control interactive and sensory features of the application through toggle switches. These options improve accessibility and personalization by enabling users to tailor the app environment according to their comfort.

- **Haptic Feedback** Enables vibration responses during button presses and gameplay interactions, creating a more tactile experience.
- **Background Music** Allows users to turn ambient music on or off while navigating lessons and practice modules.
- **Sound Effects** Controls audio cues used for correct answers incorrect responses rewards and navigation events.

The **Account Actions**section includes useful features for managing accounts. Users will be able to securely change their passwords by using Firebase Authentication services, allowing them to keep their accounts safe from the passing of time. There are also options for signing out that can safely terminate sessions on both public and private devices.

The Section has been designed using a cardbased layout with a clean, simple, and consistent typographic hierarchy; these design elements have been chosen based off of common design standards found in modern Material Designs.

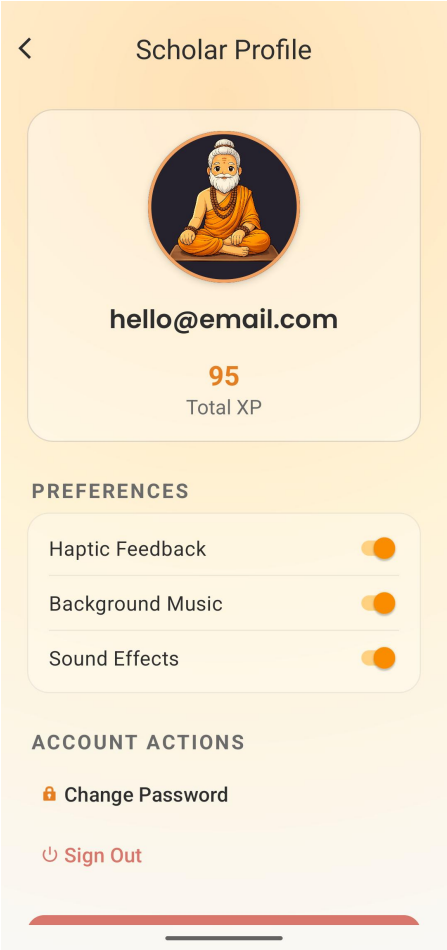


Fig. 10: Settings and Profile Page of VedaVenture

5.4 Career Mode Page

The Career Mode Page is VedaVenture's core structured and experiential educational guide facilitating user mastery of Vedic Mathematics through a full sequential curriculum. All lessons learned are compiled into a systematic course that provides guidance to users on achieving their lifelong learning objective of developing Vedic Mathematics knowledge.

Defining feature of this page: a circular progression interface of all 16 primary Vedic Sutras represented by numbered nodes and creating a visual roadmap of learner progression or mastery — this unique arrangement of Sutras also fosters a conducive atmosphere for facilitating continuous learning and motivating user progression.

Users unlock additional Sutras through both completion and consistent practice, with initial user access to only the first Sutra. All Sutras other than the first remain

locked until previous Sutras have been successfully completed by the user. Completion of each preceding unit ensures that users have learned all foundational concepts before progressing to a more advanced technique. Completion of previous units prevents random skipping of units and retains the structured logic of Vedic Mathematics as it continues to build from foundational concepts into increasingly complex techniques.

Each Sutra consists of three distinct phases:

- **Concept Introduction** The meaning and purpose, and practical application of the Sutra are explained in simplified language.
- **StepbyStep Learning** Worked examples and guided demonstrations show how the Sutra is applied to real arithmetic problems
- **Assessment and Unlocking** Users solve practice questions based on the lesson. Successful completion unlocks the next Sutra on the dial.

On this page you'll see realtime statistic information about the Student with Dynamic Learner Statistics like Total XP, Completion Rate and Current Curriculum progress These visual progress indicators will encourage students to continue learning by showing them the progress they've made or accolades they will earn.

Since the progress for the student is stored through Cloud Firebase, the student will be able to continue their path through the program without having to worry about losing any modules or earned rewards when they log in to this application after a period of absences.

To summarise, the career Mode page brings together the structure of a curriculum and gamification to make learning Vedic Maths a structured and rewarding experience for the users

Fig. 11: Career Mode Page with Interactive 16Sutra Dial

5.5 Challenge Mode Page

The Challenge Mode provides competition and skill based practice for all aspects of the application, while enhancing skills that were needed to do arithmetic quickly and accurately through interaction with a replayable game. Challenge Mode emphasizes applying Vedic Math techniques that have already been taught, as quickly as possible, within a reward based and encouraging setting.

Central to this mode of work is the intelligent random question generation engine. Instead of providing a finite amount of pre-selected questions, the system creates new arithmetic problems for users as they play the game of challenge mode. While each user plays the question generated for that user is based upon his or her personal profile providing a diverse array of problems to work on without any repeated problems The composition of each mystical question is based on two main components:

- **Difficulty Level:** Determines numerical complexity, number of digits, operation type, and response speed expected from the user.
- **Completed Sutras:** Questions are created only from Sutras that the learner has already unlocked and completed in Career Mode. This guarantees that users are tested only on techniques they have already studied.

For example someone who is a novice and has only just finished studying the first couple of Sutras could be given problems such as working with square numbers that end with 5 or finding out how to multiply using a base number, but someone who is more advanced could have to solve many different types of problems with a combination of division, simultaneous equations and/or trick methods for multiple operations.

The mode is entirely time-based, so there is a strong dependency on speed in the game. The countdown timer at the top of the screen shows how much time is left until the user runs out of time. This encourages fast decision making, high levels of mental clarity and self-confidence in pressure situations, which is particularly useful when taking aptitude tests or competing for jobs.

The upper area of the interface will include real-time statistics of the following

- Remaining Time
- Current Difficulty Tier
- Active Session Score
- Consecutive Correct Streak
- Earned XP

One of the most innovative features of this mode is the **joystickbased answer mechanism**. Instead of typing numerical answered manually, users interact with a custom circular controller positioned at the center of the screen, inspired by a joystick of video game controllers. Multiple answer choices are placed around the ring, and the learner simply slides the joystick toward the correct option. This makes game play faster, more intuitive, and significantly more enjoyable on mobile devices.

The challenge flow is as follows:

1. A random question is generated based on unlocked Sutras and difficulty level.
2. Four answer options are displayed around the joystick dial.
3. The user slides the joystick toward the chosen answer.
4. Correct answers immediately increase score and XP.
5. Consecutive correct responses build the streak multiplier.
6. A new question appears instantly until the timer ends.

The reward system is designed to encourage consistent practice habits. Users earn **Experience Points (XP)** based on accuracy, speed of response, and streak length. Daily participation helps maintain and extend long term streaks, reinforcing regular engagement with mathematics practice.

Incorrect answers do not terminate the session; instead, the system records performance data and continues generating new questions. This ensures a low pressure learning environment where mistakes become opportunities for improvement rather than punishment.

As users improve over time, the adaptive engine can automatically increase challenge intensity by reducing answer time, increasing digit size, combining multiple Sutras, or introducing more complex arithmetic patterns. Conversely, if repeated errors are detected, the system may temporarily reduce difficulty to preserve motivation and confidence.

Overall, the Challenge Mode transforms arithmetic revision into an engaging timed game where learners sharpen mental speed, gain rewards, maintain streaks, and continuously improve through unlimited randomized practice.

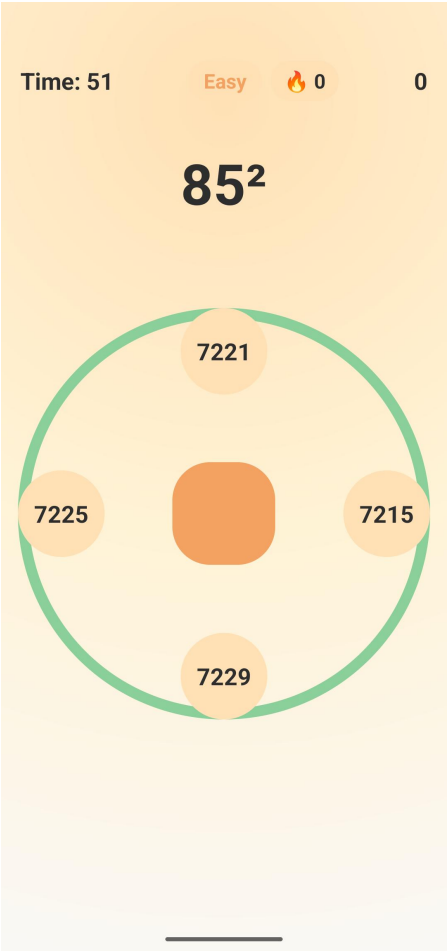


Fig. 12: Challenge Mode with Timed Randomized Questions and Joystick Based Answer Selection

5.6 Career Mode: Interactive Sutra Learning Module

A learning experience will begin as soon as a learner picks an unlocked Sutra from within the dial in Career Mode. Each of these Learning Modules (or "Modules") has been designed to convey the technique via an interactive, structured, guided, and beginner friendly way. Furthermore, these Modules contain the traditional rules of mathematics and "block" the old fashion approach to learning by being not interactive and limited to courses. Instead, they use the best of the Web to provide engaging and interactive lessons, through which learners will learn by doing, seeing, and understanding each step.

At the top of the Learning Module, the selected Sutra is shown clearly with a progress bar that indicates the status of each of the Learning Modules (i.e., how much of that Learning Module has been completed). This allows learners to keep track of where they were within that Module as well as to motivate them to complete the entire sequence of lessons covered by that Module.

The other two icons on the Learning Module screen provide contextual assistance for learners in completing the sequences covered by that Module:

- **Information Icon:** Displays the meaning of the Sutra, where it is applicable, and the general method used to solve problems with that technique.
- **Help / Question Icon:** Opens a worked demonstration of the current problem, showing the complete solution process before the learner attempts similar questions independently.

The core of the page follows a **guided instructional workflow**, where each Sutra is taught through progressive learning stages rather than a single explanation. This improves comprehension and reduces cognitive overload.

5.6.1 StepbyStep Guided Learning Process

Each problem is broken into smaller logical steps and presented one at a time. Instead of asking the learner to solve the entire question immediately, the system focuses attention on a single operation at each stage.

For example, in the case of squaring numbers ending in 5 using the *Ekadhikena Purvena* Sutra:

1. Identify the digit(s) before 5.
2. Multiply that number by its successor.
3. Append 25 to the result.
4. Combine both parts to obtain the final answer.

Each step appears individually on the screen with a dedicated prompt. The learner enters the intermediate result in the answer field and presses **Check Step**.

5.6.2 Instant Feedback Mechanism

After every response, the system immediately evaluates the learner's input and provides visual feedback:

- **Correct Intermediate Step:** A positive message such as "Correct! Now for the next step." is shown, and the lesson advances.
- **Correct Final Answer:** A completion message such as "Brilliant! Final Answer Correct." is displayed.
- **Incorrect Response:** The learner is encouraged to retry, reinforcing active participation rather than passive memorization.

This continuous feedback loop helps users identify mistakes early and learn the logic behind each stage of the solution.

5.6.3 Demo Problems and Practice Transition

The Application portion of the study guide will contain Guided Demo Problems (for example, Demo Problem 1/3) which demonstrate to students how to utilize the Sutra and apply this information to realworld scenarios before they enter into Independent Practice. These Guided Demonstration Problems will tie in theory with practical application.

Once the learner feels confident in their ability to use the Sutra, they will move to the practical portion of the module and have the opportunity to practice on their own via the "Let's Practice" section. These practice problems will require the learner to demonstrate that they can utilize the same method on their own.

5.6.4 Educational Benefits

This teaching methodology offers several learning advantages:

- Breaks complex arithmetic into manageable steps.
- Encourages conceptual understanding instead of rote memorization.
- Builds learner confidence through guided success.
- Provides immediate correction and reinforcement.
- Makes ancient Vedic techniques accessible to modern mobile learners.
- Supports selfpaced learning without teacher supervision.

Because all progress is stored using Cloud Firestore, completed lessons, solved exercises, and unlocked modules remain synchronized across sessions.

Overall, the Career Mode learning module serves as the instructional backbone of VedaVenture by converting each Sutra into an interactive tutor that teaches, tests, and motivates learners through a structured digital experience.

The images below show a general instructional delivery procedure for learning one Vedic Sutra (the sutras are shown on the first page of the workflow). Each further outlines how the 16 different Vedic Sutras have been covered according to similar specifications for students by using similar instructional frameworks to deliver their instruction. All 16 Vedic Sutras include a conceptual and stepbystep explanation (for the student) of the Sutra, a guided demonstration of the Sutra, practice problems, a system for providing feedback, and a way to unlock and continue to learn new concepts based on previous learning (progression). This, when all combined, provides an entirely unified and complete Vedic Mathematics curriculum on a single app.

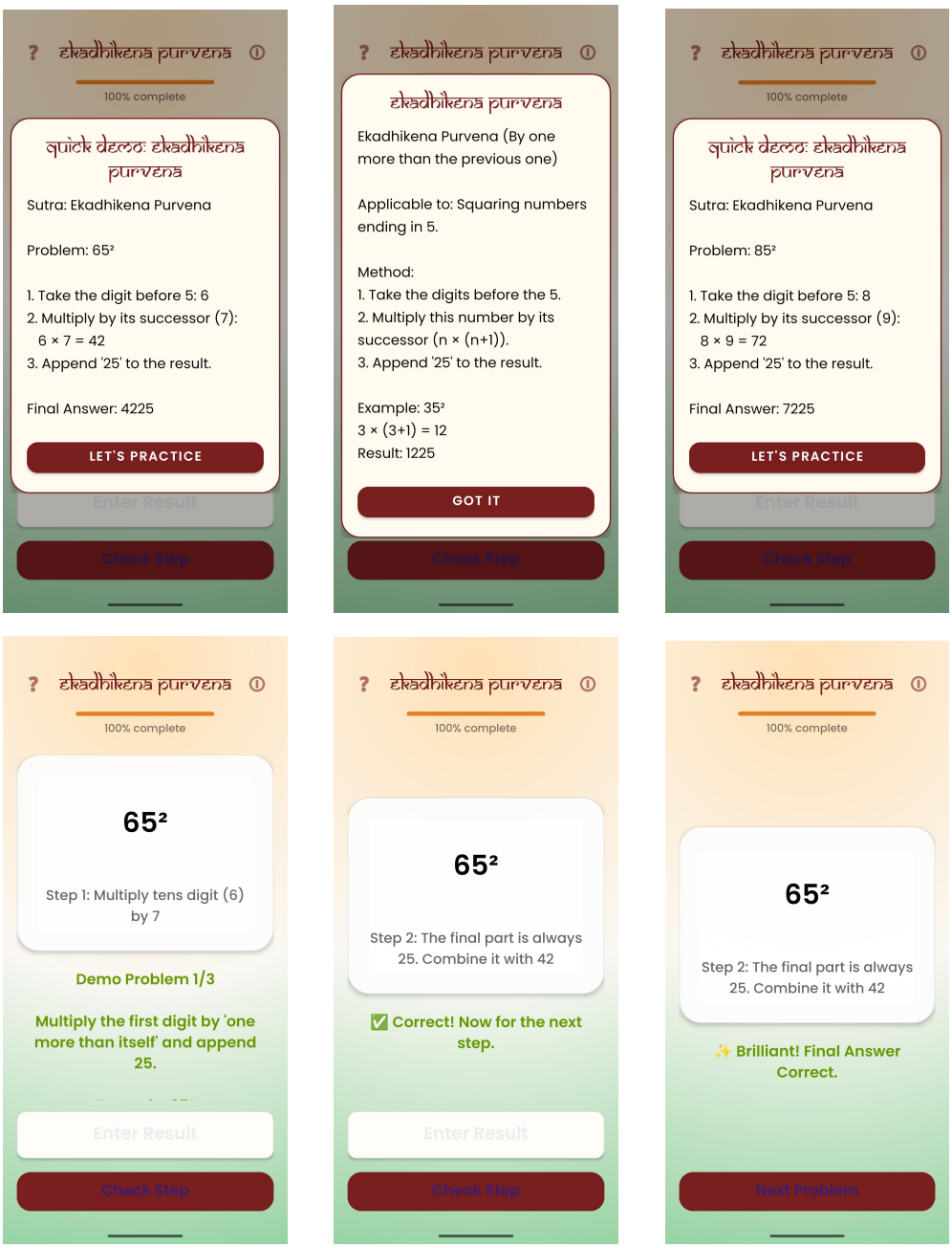


Fig. 13: StepbyStep Guided Learning Flow in Career Mode

5.7 OCRBased Problem Detection and Solving Module

One of the most innovative features of VedaVenture is the **OCRBased Problem Detection and Solving Module**, which allows users to scan realworld arithmetic questions using their device camera and instantly receive the relevant Vedic Mathematics method and solution. This feature bridges the gap between classroom problems, textbook exercises, and the digital learning environment by enabling learners to apply Vedic techniques beyond predesigned app exercises.

The module is implemented using the **Google ML Kit Text Recognition Toolkit**, which provides on device optical character recognition capabilities for faster and reliable text extractions. By performing recognition locally on the mobile device, the system reduces latency, improves privacy, and enables smoother realtime interaction without cloud dependence.

5.7.1 Working Procedure

The OCR workflow is divided into multiple stages:

1. **Camera Capture:** The user opens the OCR Scanner page and points the camera toward a mathematical expression printed on paper, displayed on a screen, or handwritten very clearly.
2. **Region Detection:** The scanner identifies the visible problem area and highlights the detected expression using a bounding box overlay.
3. **Text Recognition:** ML Kit extracts the characters, symbols, and numbers from the image and converts them into machine readable text.
4. **Expression Parsing:** The recognized text is analyzed to identify operators such as addition, subtraction, multiplication, powers, or division.
5. **Sutra Mapping Engine:** Based on the structure of the problem, the system predicts the most suitable Vedic Sutra for solving it.
6. **Interactive Solution Output:** The learner is asked whether they would like to solve the problem using the suggested Sutra. If accepted, the app displays the answer and explains the stepbystep method.

5.7.2 Automatic Sutra Detection Logic

The OCR module does not merely read numbers; it also performs rulebased classification to connect detected problems with learned Sutras. Examples include:

- Numbers ending in 5 with square notation (e.g., 85^2) $\rightarrow$ *Ekadhikena Purvena*
- Multiplication of general multidigit numbers (e.g., 3×55) $\rightarrow$ *Urdhva Tiryakbhyam*
- Numbers close to powers of 10 (e.g., 98×97) $\rightarrow$ *Nikhilam*
- Division or transposition patterns $\rightarrow$ *Paravartya Yojayet*

This intelligent mapping allows learners to understand not only the answer, but also *which* Vedic technique should be applied in different scenarios.

5.7.3 Educational Benefits

The OCR module significantly extends the usefulness of the platform by allowing spontaneous practice from external sources. Instead of manually entering every question, users can simply scan and learn instantly.

Key benefits include:

- Converts textbook and worksheet questions into interactive learning opportunities.
- Helps learners identify the correct Sutra for unfamiliar problems.
- Provides immediate worked solutions and conceptual guidance.
- Encourages realworld application of Vedic Mathematics outside the app.
- Reduces typing effort and improves accessibility on mobile devices.
- Creates a smart tutoring experience using machine learning assistance.

5.7.4 Technical Advantages of ML Kit Integration

Using Google ML Kit provides several implementation advantages:

- Fast on device inference with low latency.
- Works offline for majority of recognition tasks.
- Supports camera stream integration in Android applications.
- Reliable recognition of printed numeric expressions.
- Easily integrates with Java/Kotlin Android projects.

Overall, the OCR Problem Detection and Solving Module transforms the smartphone camera into an intelligent mathematics assistant, enabling users to scan, understand, and solve arithmetic problems through the power of Vedic Mathematics and machine learning.

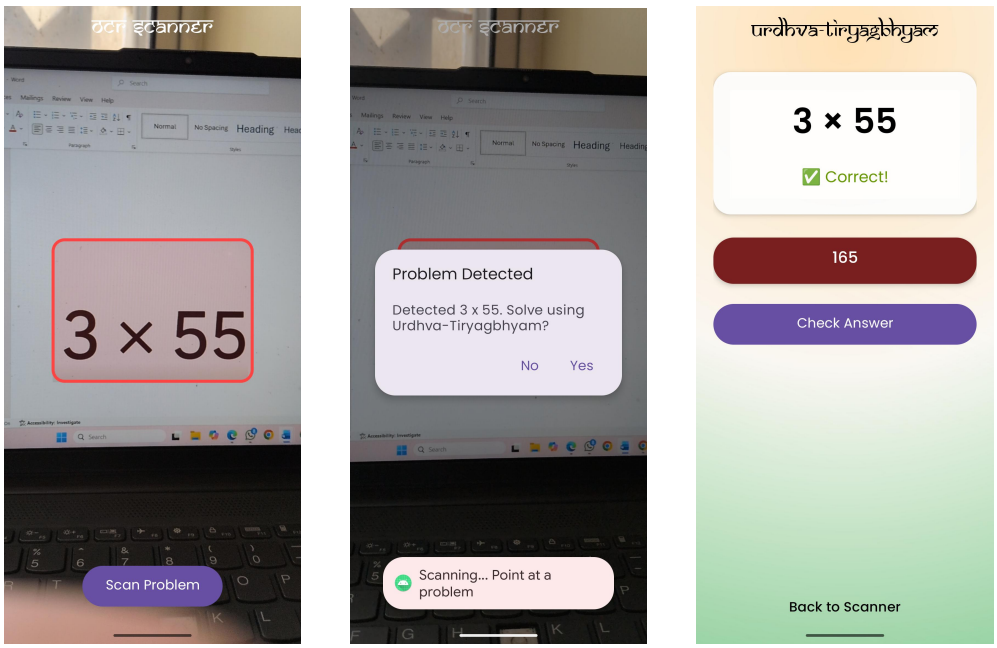


Fig. 14: OCR Scanner: Problem Detection, Recognition, and Solution Output

6 Software Requirement

6.1 Functional Requirements

- The system shall allow new users to register with a valid email and password.
- The system shall authenticate registered users and grant access to the application upon successful login.
- The system shall display a Career Mode with a list of Vedic Sutras, indicating locked and unlocked status for each.
- The system shall allow users to unlock the next Sutra only after successfully completing the assessment phase of the current Sutra.
- The system shall present stepbystep instructional content for each Sutra in Career Mode.
- The system shall present practice questions for each Sutra, with a hint system available for each question.
- The system shall provide a Challenge Mode with timebound, dynamically generated Vedic arithmetic questions.
- The system shall implement a joystickbased directional input for answering Challenge Mode questions.
- The system shall award XP to users based on correctness, speed, and difficulty of answers.
- The system shall maintain and display a daily streak counter, resetting after 24 hours of inactivity.

- The system shall record and display the user's best streak.
- The system shall adapt question difficulty in Challenge Mode based on user accuracy and response time.
- The system shall provide an OCR scanner that captures an arithmetic question from an image and returns the relevant Vedic Sutra and stepbystep solution.
- The system shall persist all user data, including XP, streaks, Sutra progress, and difficulty level, to Firebase Firestore.
- The system shall synchronize user data in real time across devices.

7 Conclusion

The mobile application has resolved the issue of low motivation and engagement associated with traditional methods of teaching arithmetic. By using Vedic maths on an gamified and adaptive mobile platform, it enhances both user motivation and their ability to perform simple calculations. By having structured Career Learning, adaptive Challenge Testing and realtime OCR problem-solving together this application can also provide users with a complete ecosystem that helps them develop their mathematical skills. Potential enhancements to this application may include adding additional OCR capabilities to support functions such as solving complex algebraic equations, using advanced analytics to observe where users are experiencing difficulty, and offering a multiplayer leaderboard to further utilise the gamification of the application